Proof

The k-permutations of the set can be obtained by forming the $C(n,k)$ k-combinations of the set, and then ordering the elements in each k-combination, which can be done in $P(k,k)$ (i.e., $k!$) ways. Consequently,

$$P(n,k) = C(n,k) \cdot P(k,k)$$

This implies that

$$C(n,k) = \frac{P(n,\ k)}{P(k,k)} = \frac{n!/(n-k)!}{k!} = \frac{n!}{k!\,(n-k)!}$$